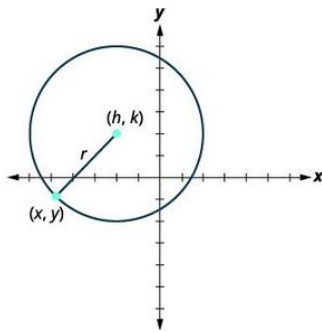


Circle

A circle is a closed two-dimensional figure in which the set of all the points in the plane is equidistant from a given point called the centre'. The circle formula in the plane is given as:

$$(x - h)^2 + (y - k)^2 = r^2$$

Where (x, y) are the coordinate points (h, k) is the coordinate of the centre of a circle and r is the radius of a circle.



General Form of the Equation of a Circle

The general equation of the circle is represented by:

$$x^2 + y^2 + 2gx + 2fy + c = 0, \text{ for all values of } g, f, \text{ and } c.$$

Adding $g^2 + f^2$ on both sides of the equation gives,

$$x^2 + 2gx + g^2 + y^2 + 2fy + f^2 = g^2 + f^2 - c$$

Since $(x+g)^2 = x^2 + 2gx + g^2$ and $(y+f)^2 = y^2 + 2fy + f^2$, substituting the values in equation (1), we have

$$(x+g)^2 + (y+f)^2 = g^2 + f^2 - c$$

Comparing (2) with $(x-h)^2 + (y-k)^2 = a^2$, where (h, k) is the center and 'a' is the radius of the circle.

$$h = -g, k = -f$$

$$a^2 = g^2 + f^2 - c$$

Therefore,

$x^2 + y^2 + 2gx + 2fy + c = 0$, represents the circle with centre $(-g, -f)$ and radius equal to $a^2 = g^2 + f^2 - c$.

- If $g^2 + f^2 > c$, then the radius of the circle is real.

- If $g^2 + f^2 = c$, then the radius of the circle is zero, which tells us that the circle is a point that coincides with the center. Such a type of circle is called a point circle
- $g^2 + f^2 < c$, then the radius of the circle becomes imaginary. Therefore, it is a circle having a real center and an imaginary radius.

Polar Equation of a Circle

To find the polar form of the equation of a circle, replace the value of $x = r \cos \theta$ and $y = r \sin \theta$,

Here, $x^2 + y^2 = a^2$.

Hence, we get;

$$(r \cos \theta)^2 + (r \sin \theta)^2 = a^2$$

$$r^2 \cos^2 \theta + r^2 \sin^2 \theta = a^2$$

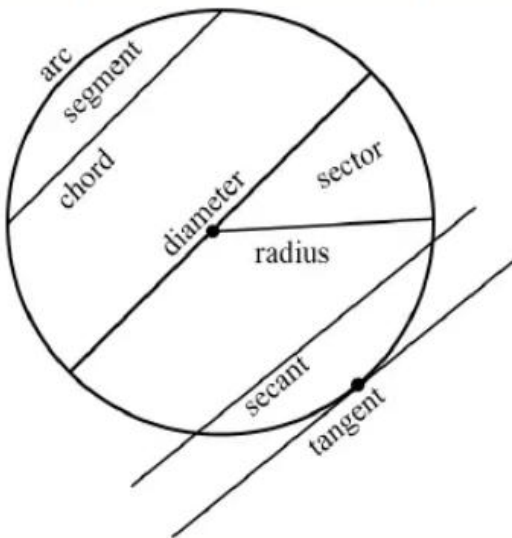
$$r^2 (\cos^2 \theta + \sin^2 \theta) = a^2$$

$$r^2 (1) = a^2 \text{ [Using trigonometry identity]}$$

$$r = a$$

is the polar equation of a circle with radius a and center at the origin $(0,0)$.

BASIC CIRCLE TERMINOLOGY



Radius:

A line from the center to any point on the Circumference of the circle.

Diameter:

A line passing through the center of the circle. It is double the length of the radius.

Chord:

A line with end-points on the circumference.

Secant:

A line passing through two points on the circle.

Tangent:

A line touching the circle at only one point.

** The equation of the circle whose diameter is the line joining the points $(x_1, y_1), (x_2, y_2)$ is

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

** The equation of the tangent to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at (x_1, y_1) is

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

** The equation of the pair of tangents drawn from a point (x_1, y_1) to the circle $x^2 + y^2 = a^2$ is

$$(x^2 + y^2 - a^2)(x_1^2 + y_1^2 - a^2) = (xx_1 + yy_1 - a^2)^2$$

Art. 70. Length and mid point of the chord intercepted by the circle $x^2 + y^2 = a^2$ on the line $y = mx + c$.

The line joining centre O to the middle point C of a chord of a circle is perpendicular to the chord whose equation is $y = mx + c$

Hence the equation of OC is $y - 0 = -1/m x$ or, $my + x = 0$... (1)

Solve (1) and (2), the co-ordinates of the middle point C of the chord AB is $-mc/(1 + m^2), c/(1 + m^2)$... (2)

Again OC = perpendicular distance of C (0,0) from AB ... (3)

$OC = \frac{c}{\sqrt{1 + m^2}}$ and $CA^2 = a^2 - c^2/(1 + m^2)$

Length of chord AB = 2AC.

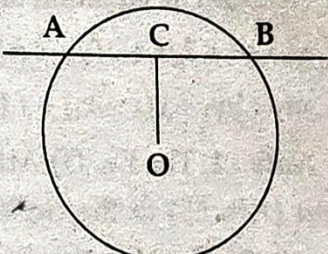
$$= 2\sqrt{\left\{\frac{a^2(1 + m^2) - c^2}{(1 + m^2)}\right\}}$$


Fig : 30

The equation of the pair of tangents to a circle

The equation for a pair of tangents from an external point (x_1, y_1) to a circle is given by the formula

$$SS_1 = T^2$$

S is the equation of the circle,

S_1 is the result of substituting the point's coordinates into S ,

T is obtained by a specific linear combination of the coordinates. For a general circle

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Where,

$$S = x^2 + y^2 + 2gx + 2fy + c,$$

$$S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c,$$

$$T = xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c$$

Examples:

Ex. 1. Find the equation of the circle passing through the three points $(-3, 2)$, $(1, 7)$ and $(5, -3)$

The general equation of all circles passing through the first two points $(-3, 2)$, $(1, 7)$ is
 $(x+3)(x-1) + (y-2)(y-7) = A \{(x+3)(y-7) - (x-1)(y-2)\} \dots (1)$

Since the circle passes through the third point $(5, -3)$, we have $8.4 + (-5)(-10) = A \{(8(-10) - 4(-5))\}$ or, $A = -30/41$

Thus from (4), we have

$$30(x^2 + y^2 + 2x - 9x + 11) = -41(-5x + 4y - 23)$$

$$\text{or, } 30(x^2 + y^2) - 145x - 106y - 613 = 0$$

which is the required equation.

Ex. 2. Find the pole of the straight line $2x - y = 6$ w. r. to the circle $5x^2 + 5y^2 = 9$

Let $P(x_1, y_1)$ be the co-ordinates of the pole.

The equation to the polar of $P(x_1, y_1)$ with respect to the circle

$$5x^2 + 5y^2 = 9 \text{ is } 5xx_1 + 5yy_1 = 9 \dots (1)$$

$$\text{This polar is identical with the line } 2x - y = 6 \dots (2)$$

Now comparing the co-efficients of x, y from (1) and (2) we have. $5x_1/2 = -5y_1/1 = 9/6$,
 $x_1 = 3/5, y_1 = -3/10 \therefore (3/5, -3/10)$ Ans.

Ex. 6. Show that the equation to the pair of tangents drawn from the origin to the circle
 $x^2 + y^2 + 2gx + 2fy + c = 0$

$$\text{is } (gx + fy)^2 + c(x^2 + y^2)$$

The equation to the pair tangents is $SS_1 = T^2$

$$\text{or, } (x^2 + y^2 + 2gx + 2fy + c)c = x[x \cdot 0 + y \cdot 0 + g(x+0) + f(y+0) + c]^2$$

$$\text{or, } c(x^2 + y^2 + 2gx + 2fy + c) = (gx + fy + c)^2$$

$$\text{or, } c(x^2 + y^2) + c(2gx + 2fy + c) = (gx + fy)^2 + 2(gx + fy)c + c^2$$

$$\text{or, } c(x^2 + y^2) = (gx + fy)^2$$

Ex. 8. Find the equation of the circle inscribed in the triangle determined by the lines.

$$2x - 3y + 21 = 0 \quad \dots \quad (i)$$

$$3x - 2y + 20 = 0 \quad \dots \quad (ii)$$

$$2x + 3y + 9 = 0 \quad \dots \quad (iii)$$

The centre of the circles lies at the point of intersection of the bisectors of the interior angles of the triangle. Let (h, k) be the co-ordinates of the centre of the incircle. Bisector of angle (i) and (ii) is

$$\frac{2h - 3k + 21}{\sqrt{4 + 9}} = + \frac{3h - 2k - 6}{\sqrt{9 + 4}} \quad \text{or, } h + k + 3 = 0 \quad (iv)$$

Bisector of angle (i) and (iii) is

$$\frac{2h - 3k + 21}{\sqrt{4 + 9}} = + \frac{2h + 3k + 9}{\sqrt{4 + 9}} \quad \text{or, } 6k - 12 = 0 \quad \text{or, } k = 2 \quad (v)$$

From (iv), $h = -1$. Co. ordinates of centre $(-1, 2)$

$$\text{Radius of the circle, } = \frac{2(-1) + 3(2) + 9}{\sqrt{13}} = \sqrt{13}$$

Hence equation to circle is $(x + 1)^2 + (y - 2)^2 = (\sqrt{13})^2$, or, $x^2 + y^2 + 2x - 4y - 8 = 0$

Ex. 11. The circle $x^2 + y^2 - 2x + 4y - 20 = 0$ meets the straight line $2x - y = 2$ at A and B. Find the equation of the straight line joining A and B with the origin and obtain the angle between them.

The equation of the straight lines OA and OB is obtained by making the equation of the circle $x^2 + y^2 - 2x + 4y - 20 = 0$

homogeneous with the line $2x - y = 2$ or, $\frac{1}{2}(2x - y) = 1$

The joint equation is $x^2 + y^2 - 2x \cdot \frac{1}{2}(2x - y) + 4y \cdot \frac{1}{2}(2x - y) - 20 \left\{ \frac{1}{2}(2x - y) \right\}^2 = 0$
or, $x^2 + y^2 - 2x^2 + xy + 4xy - 2y^2 - 5(4x^2 + y^2 - 4xy) = 0$
or, $21x^2 + 6y^2 - 25xy = 0$

And the angle between the straight lines is $\tan 2\phi = \frac{2\sqrt{(h^2 - ab)}}{a + b}$

$$\text{or, } \tan 2\phi = \frac{2\sqrt{(625/4 - 21.6)}}{21 + 6} \quad \therefore 2\phi = \tan^{-1} \frac{11}{27}$$

Exercise:

1. Find the pole of the lines $2x - y + 10 = 0$ w.r.t $x^2 + y^2 - 7x - 5y - 1$. Ans: $(3/2, -3/2)$
2. Find the equation of the pair of tangents from the points $(5,5)$ to the circle $x^2 + y^2 - 6x - 6y + 14 = 0$.

3. Find the equation of the circle inscribed in the triangle formed by the lines $2x - 3y + 21 = 0$, $3x - 2y - 6 = 0$, $2x + 3y + 9 = 0$. Ans: $x^2 + y^2 + 2x - 4y - 8 = 0$
4. Find the equation of the circle passing through the points $(1,3)$, $(2,-1)$, $(-1,1)$.
Ans: $5x^2 + 5y^2 - 11x - 9y - 12 = 0$.